Multiple Probabilities of Direction: A Rationale for Prior-odds Adjustment

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Abstract

In psychological research, Bayesian analyses are often summarized using posterior indices and then interpreted with conventional cutoffs across many parameters. The probability of direction (*pd*) is one such index, defined as the posterior probability that a parameter is greater or smaller than a null value. When *pd* is thresholded and used repeatedly, multiplicity becomes consequential but is rarely handled explicitly. Here, we examine the operating characteristics of *pd* under repeated use by leveraging its correspondence with the frequentist *p*-value under common settings. Simulations show that extreme *pd* can arise by chance and that misleading conclusions become more frequent as the number of tests increases. We further express *pd* in terms of posterior odds and propose prior-odds adjustments that provide multiplicity control while preserving Bayesian coherence. To support practical implementation, we provide an R package that applies the proposed prior-odds corrections.

Keywords: probability of direction, multiple testing, prior-odds

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Over the past two decades, the proportion of articles reporting Bayesian analyses in psychology has increased considerably (Jevremov & Pajić, 2024; Pfadt et al., 2025; Van De Schoot et al., 2017). Unfortunately, this expansion has been accompanied by practices reminiscent of the "null ritual" in frequentist statistics, where widespread use was associated with automatic application rather than principled reasoning (Cohen, 1994; Gigerenzer, 2004). For instance, researchers sometimes treat Bayes factor thresholds (e.g., BF > 3) as dichotomous decision rules or claim that an effect "exists" because the posterior probability of a positive effect exceeds a conventional benchmark such as 95% (Heck et al., 2023; Hoijtink et al., 2016; Kelter, 2023; Kruschke, 2018; Tendeiro et al., 2024; Tendeiro & Kiers, 2019). While such summaries are convenient, their use as default cutoffs encourages all-or-none interpretations. When these binary judgments are applied repeatedly across many parameters, the issue of multiplicity arises.

In the frequentist framework, multiplicity is typically framed as Type I error (false-positive) problem: if each individual test is conducted at level α, the probability of at least one false positive increases with the number of tests. Classical approaches therefore control either the familywise error rate (the probability of at least one Type I error) or the false discovery rate (the expected proportion of false positives among rejected hypotheses) [**REFs**]. However, multiplicity concerns extend beyond false positives. When researchers examine many parameters and highlight the most extreme estimates, selection effects can produce exaggerated magnitudes (Altoè et al., 2020; Gelman & Tuerlinckx, 2000; Zwet & Cator, 2021). Bayesian shrinkage methods address these distortions by pulling extreme estimates toward more plausible values. Such shrinkage can arise through hierarchical models with partial pooling (Berry & Hochberg, 1999; Gelman et al., 2012, 2013; Kruschke, 2011) and/or through informative priors that reduce overestimation in low signal-to-noise settings (Dawid, 1994; Zwet & Gelman, 2022).

Regularization improves estimation, but it does not by itself calibrate repeated threshold-based decisions across many parameters. If the same cutoff is applied to many parameters (e.g., declaring an effect whenever posterior probability exceeds 0.95), the chance of at

least one spurious finding can remain inflated, even if shrinkage reduces it. Bayesian approaches that directly target this issue typically rely on either simultaneous credible statements (Box & Tiao, 1992; Gelman & Tuerlinckx, 2000; Pruzek, 2016; Schnell et al., 2020) or priors on the hypothesis/model space that scale with the number of tested effects (Jeffreys, 1938; Scott & Berger, 2006; Scott & Berger, 2010; Westfall et al., 1997). Nevertheless, psychological researchers often conflate selection effects that distort estimates with the distinct problem of making many simultaneous decisions. As a result, shrinkage is sometimes treated as sufficient for multiplicity control (Chen et al., 2020; Erčulj & Štrumbelj, 2015; Hogeveen et al., 2022; Limbachia et al., 2021; Mathar et al., 2022).

Other justifications invokes the *subjectivist* interpretation of Bayesian inference, which treats posterior probabilities as rational degrees of personal belief (De Finetti, 2017; Edwards et al., 1963; Lindley, 1972; Rozeboom, 1960). However, this interpretation fits poorly with common psychological practice, where researchers typically apply default software priors (often weakly informative) rather than priors encoding substantive domain knowledge (Smid & Winter, 2020; Van De Schoot et al., 2017). When default priors are combined with sufficient data, Bayesian and frequentist procedures often converge to similar conclusions (Albers et al., 2018; Bernardo & Smith, 1994; Gelman et al., 2013). This convergence makes it both natural and useful to evaluate the frequentist operating characteristics of Bayesian decision rules under multiplicity.

Accordingly, we adopt a *falsificationist* perspective, which treats Bayesian procedures as tools to be calibrated rather than as solely expressions of belief (Box, 1980; Gelman & Hennig, 2017; Gelman & Shalizi, 2013; Mayo, 2018; Rubin, 1984). Here, we focus on the Probability of Direction (*pd*), a Bayesian index that can be mapped directly to the frequentist *p*-value in common settings (Berger & Sellke, 1987; Casella & Berger, 1987; Makowski et al., 2019; Marsman & Wagenmakers, 2017). This correspondence allows us to evaluate *pd*'s behavior under repeated use and to compare Bayesian and frequentist approaches to multiplicity control on common ground.

The paper proceeds in four steps. First, we introduce *pd* and make explicit its connection to the frequentist *p*-value, which provides a reference point for calibration. Second, we

characterize *pd*'s null distribution and examine its multiple-testing behavior through simulations that quantify false-positive rates. Third, we relate *pd* to directional Bayes factors and formalize the role of prior odds. Finally, we present prior-odds adjustments for multiplicity following Jeffreys (Jeffreys, 1938; see also Westfall et al., 1997), discuss implementation under dependence, and provide guidance for reporting both raw and multiplicity-adjusted *pd*.

Probability of Direction and *p*-value

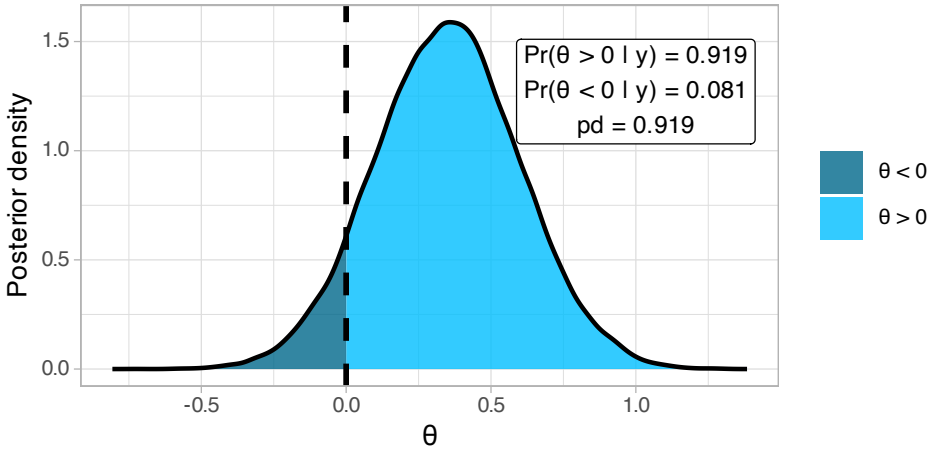
The Probability of Direction (*pd*) for continuous distributions can be defined as the maximum between the posterior probabilities that a parameter is greater or smaller than the null value:

$$pd = \max(\Pr(\theta < \theta_{null} \mid y),\ \Pr(\theta > \theta_{null} \mid y)).$$

As proposed and popularized in applied Bayesian workflows, *pd* offers an intuitive summary of directional evidence by quantifying the proportion of posterior that lies on the dominant side (Makowski et al., 2019); it ranges from 0.50 (no directional preference) to 1 (all on one side).

Figure 1

Posterior distribution with shaded directional posterior area. In this example, the probability of direction (pd) is the greater posterior probability on either side of 0.



Because *pd* is a functional of the posterior, it inherits dependence on the full model specification (likelihood and prior), so its long-run error behavior is not a property of *pd* "in isolation" but of the overall Bayesian procedure that generates the posterior. A convenient starting

point for calibration is the mapping between pd and the frequentist p-value. Under flat (or sufficiently diffuse symmetric) priors in typical hypothesis-testing scenarios (e.g., testing means), pd is simply the posterior on the dominant side of θ_{null}, which corresponds to the complement of the one-sided p-value (Casella & Berger, 1987; Marsman & Wagenmakers, 2017):

$$pd = \max(Pr(\theta < \theta_{null} \mid y), Pr(\theta > \theta_{null} \mid y)) = 1 - p,$$

with the two-sided correspondence being $p_{two-sided} = 2p = 2(1 - pd)$.

Probability of Direction as a test of existence and direction

The pd can be embedded in a testing framework, where thresholding $pd > c$ yields a claim that an effect exists *and* has a particular sign. For instance, when $pd > c$ supports $\theta > 0$, the posterior probability assigned to nonpositive values is bounded by

$$\Pr(\theta \leq 0 \mid y) < 1 - c$$

(analogously, when $pd > c$ supports $\theta < 0$, $\Pr(\theta \geq 0 \mid y) < 1 - c$). This decision rule does more than compare H_+ versus H_-, it also excludes 0 by ensuring that posterior mass on the null-or-opposite-sign region remains below $1 - c$. Since $\Pr(\theta \leq 0 \mid y) < 1 - c$, zero must lie below the $(1 - c)$-quantile of the posterior distribution. Equivalently, the $(1 - c)$-quantile is strictly positive. As a result, the central credible interval (bounded by the $(1 - c)$ and c quantiles) excludes 0 (see Figure 2).

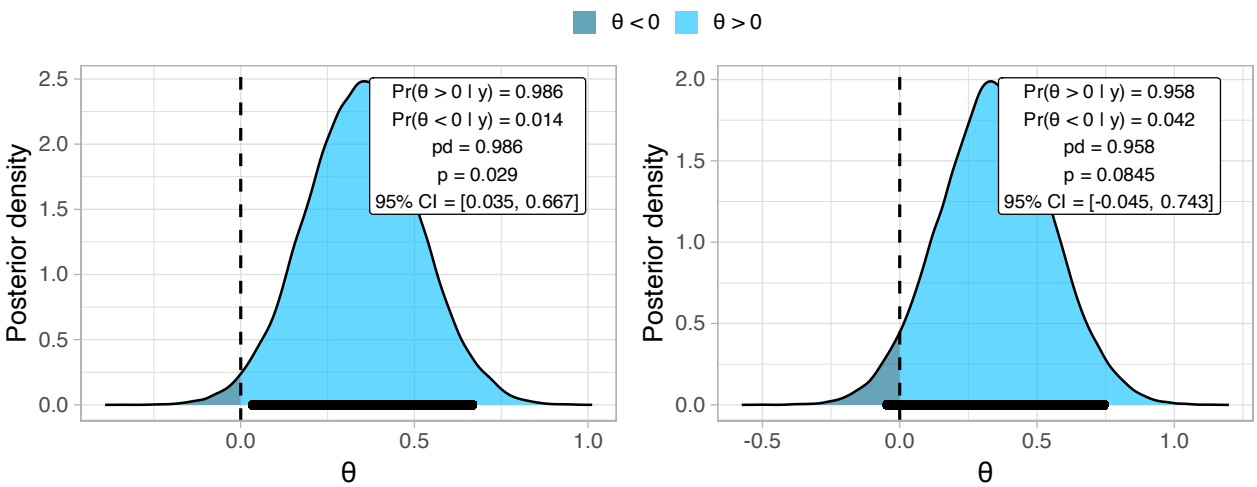
Pure directional testing conditions on the assumption that $\theta \neq 0$ and evaluates only the sign of the effect by comparing H_+ versus H_-; $H_0 : \theta = 0$ is not among the hypotheses under consideration. In contrast, here we explicitly include H_0 and separates decisions about *existence* (excluding zero) from decisions about direction (positive vs negative). Error types follow directly from this distinction: a Type I error occurs when the true parameter lies in H_0 but H_0 is rejected in favor of H_+ or H_-.

Simulation setup

In our simulations we generate data under the global null (all $\theta = 0$) and we compute $\Pr(pd > c \mid \theta = 0)$. Operationally, this quantifies how often the procedure falsely excludes zero

Figure 2

Posterior distributions illustrating the relationship between the probability of direction (pd) and credible-interval (CI), using $c = 0.975$ and $\theta_{null} = 0$. ***Left:*** *A narrow posterior whose 95% CI excludes 0.* ***Right:*** *A wider posterior whose 95% CI overlaps 0.*



from the credible region, that is, incorrectly concludes that an effect exists (with a particular sign) when the true parameter is zero. Specifically, we conduct a Monte Carlo simulation in which, for each condition, we generated families of size $m \in \{1, 2, 5, 10, 25\}$, with $n \in \{30, 50, 100\}$ observations per dataset, and repeated the procedure 10,000 times. In every replication, data were generated under the global null: each dataset had true mean 0 and unit variance, $y \sim \mathcal{N}(0, 1)$. Each dataset was analyzed using a conjugate normal model with known variance $\sigma^2 = 1$ and prior $\theta \sim \mathcal{N}(0, \tau^2)$, where $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$ spans skeptical to flat priors. We drew posterior samples of θ from the resulting normal posterior and computed *pd* as $pd = \max(\Pr(\theta > 0 \mid y), \Pr(\theta < 0 \mid y))$ using these draws. For the frequentist comparison, we conducted two-sided one-sample z-tests.

Prior influence

Under the null hypothesis, two-sided p-values are uniformly distributed on $[0, 1]$, and because *pd* can be mapped to the p-value, this implies a corresponding null reference behavior for *pd* on $[0.5, 1]$ in the continuous case. However, because the *pd* is a posterior probability about the sign of θ, its null distribution is not determined by the likelihood alone: it can shift depending on

how strongly the prior concentrates probability near 0 (i.e., how readily the prior "expects" near-null effects). Showing this prior dependence at $m = 1$ makes the target of later multiplicity analyses explicit: even before any multiplicity is introduced, a fixed cutoff such as $pd > 0.975$ does not have a universal "null meaning" across priors and sample sizes (see Figure 3). Put differently, the first simulation establishes the baseline calibration of "extreme pd" under the chosen model. With this baseline in hand, multiplicity can be introduced as a second, distinct mechanism: even if each individual test has only a small chance of producing an extreme pd under H_0, the probability that at least one of $pd_1, \ldots, pd_m$ is extreme increases with m. To make this separation explicit, the simulation extends from $m = 1$ to $m = 20$ under the global null and summarizes familywise behavior by computing the probability of at least one $pd > 0.975$ across the family of tests (Figure 4).

Figure 3

Null reference distributions for p-values and probability of direction. The ***panel A*** *shows the histogram of two-sided one-sample z-test p-values from 10,000 replications under the global null ($\theta = 0$), with binwidth 0.025. The* ***panel B*** *shows the corresponding empirical distributions of the probability of direction (pd) computed from a conjugate normal posterior, stratified by sample size ($n = 30, 50, 100$; rows) and prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$; columns). The vertical dashed line marks the commonly used evidential cutoff (pd = 0.975), which corresponds to a two-sided p-value of 0.05 under the standard conversion $p = 2(1 - pd)$.*

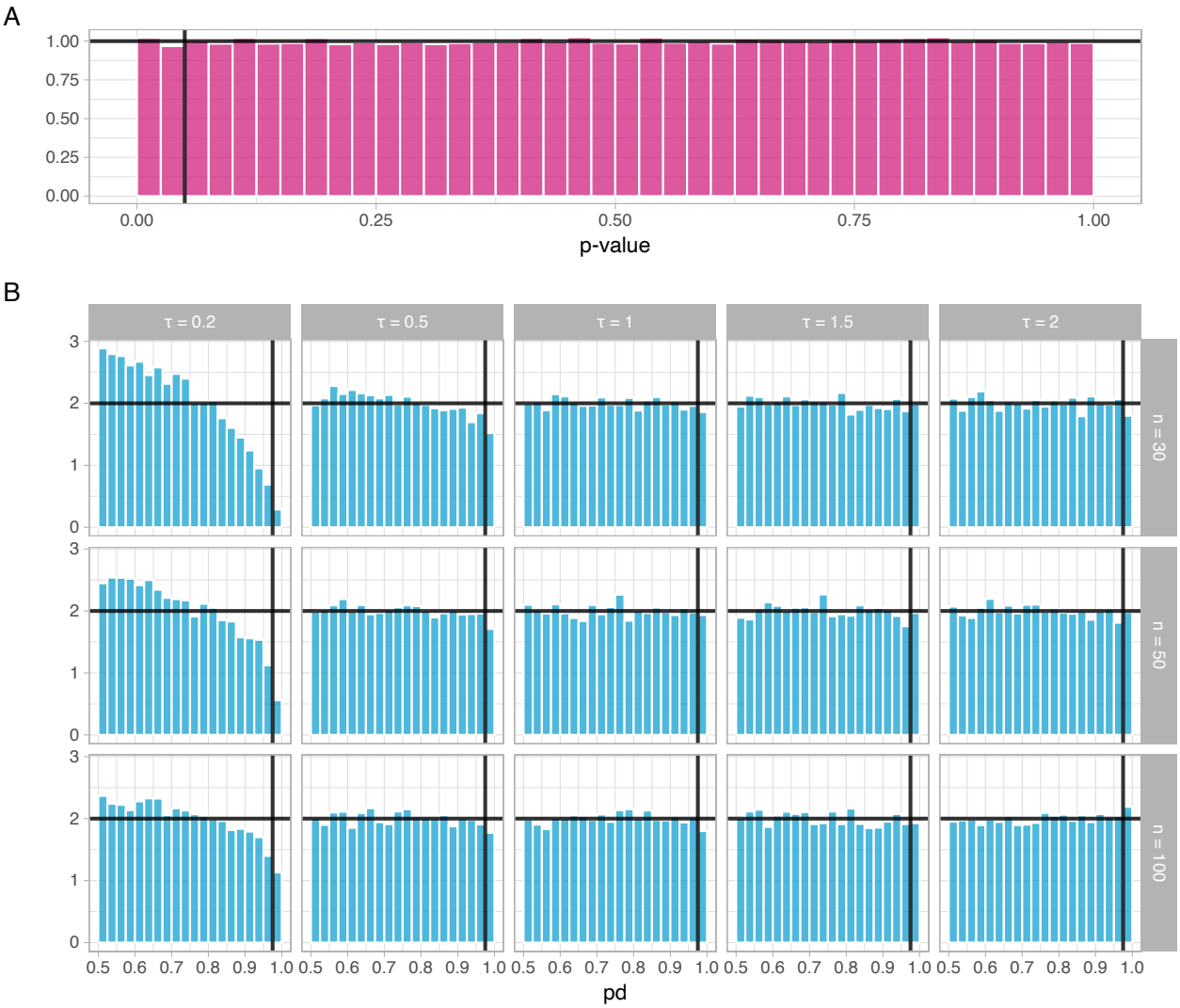
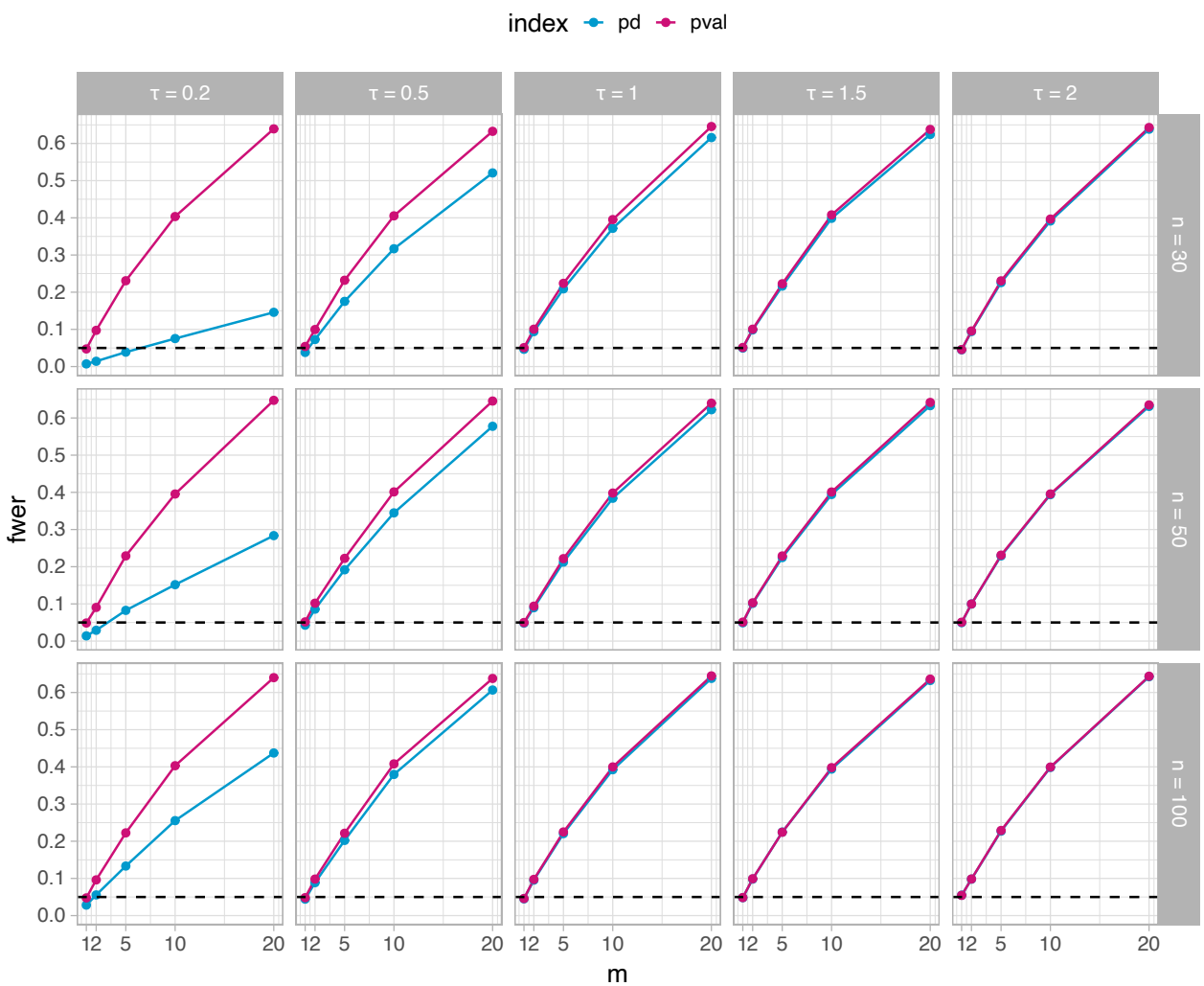


Figure 4

Familywise error rates under the global null for p-values and probability of direction. The figure shows the estimated familywise error rate (fwer), defined as the probability of at least one false positive across m tests, as a function of the number of tests (m), based on 10,000 replications under the global null ($\theta = 0$ for all tests). The pink curve corresponds to declaring at least one discovery when at least one two-sided one-sample z-test p-value < 0.05, whereas the blue curve corresponds to declaring at least one discovery when at least one pd > 0.975 under the conjugate normal Bayesian model. Rows vary the per-test sample size ($n = 30, 50, 100$), and columns vary the prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$).



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Probability of Direction, posterior odds, and prior-odds adjustment

From posterior probabilities to posterior odds

The probability of direction (pd) is defined as the maximum of the two posterior tail probabilities around a null value θ_{null}. For notational simplicity, we set $\theta_{\text{null}} = 0$ throughout this section. To connect pd to an evidential interpretation, consider the directional hypotheses $H_+ : \theta > 0$ and $H_- : \theta < 0$. If the prior for θ is continuous and symmetric about 0, then $\Pr(H_+) = \Pr(H_-) = 1/2$. Without loss of generality, assume the posterior favors H_+, so that $pd = \Pr(H_+ \mid y) = \Pr(\theta > 0 \mid y)$. The complement probability is then $1 - pd = \Pr(\theta \leq 0 \mid y)$; under a continuous symmetric prior and posterior, $\Pr(\theta = 0 \mid y) = 0$ and therefore $1 - pd = \Pr(\theta < 0 \mid y) = \Pr(H_- \mid y)$ (the "$\leq$" notation simply makes the boundary inclusion explicit). Hence the posterior odds for the favored direction are

$$\frac{pd}{1 - pd} = \frac{\Pr(H_+ \mid y)}{\Pr(H_- \mid y)}.$$

By Bayes' rule, these odds factorize into a Bayes factor and prior odds,

$$\frac{\Pr(H_+ \mid y)}{\Pr(H_- \mid y)} = \text{BF}_{+/-} \times \frac{\Pr(H_+)}{\Pr(H_-)},$$

where $\text{BF}_{+/-} = p(y \mid H_+)/p(y \mid H_-)$ is the directional Bayes factor. Under prior symmetry, $\Pr(H_+)/\Pr(H_-) = 1$, so $\frac{pd}{1-pd} = \text{BF}_{+/-}$ (Casella & Berger, 1987; Marsman & Wagenmakers, 2017). Thus, the odds $pd/(1 - pd)$ computed from posterior draws can be interpreted directly as evidence favoring $\theta > 0$ over $\theta \leq 0$.

Multiplicity adjustment via post-hoc prior-odds

When assessing a single directional claim under a symmetric prior centered at the null, prior odds equal 1, encoding prior indifference between positive and negative effects. However, when m hypotheses are assessed simultaneously, assigning prior probability $\frac{1}{2}$ to each directional claim implies a joint prior that rapidly concentrates probability on the event that *at least one* effect

departs from 0. For instance, with $m = 10$ the implied prior probability that all effects are "not in the favored direction" is $0.5^{10} \approx 0.001$, which is often unrealistically low.

A principled way to address multiplicity is to decrease the prior odds assigned to any particular directional claim as a function of m, while leaving the the Bayes factor component unchanged. Because the analysis is symmetric in the sign of θ and pd is defined as the posterior probability of the dominant direction, the same adjustment can be applied regardless of whether the posterior favors $\theta > 0$ or $\theta < 0$.

Concretely, let $\Pr(H_0)$ and $\Pr(H_1)$ denote the multiplicity-adjusted prior probabilities assigned to the favored direction and its complement, respectively (with $\Pr(H_0) + \Pr(H_1) = 1$). The adjusted posterior odds are then

$$\frac{pd_{\text{adj}}}{1 - pd_{\text{adj}}} = \frac{pd}{1 - pd} \times \frac{\Pr(H_1)}{\Pr(H_0)},$$

so that, equivalently,

$$pd_{\text{adj}} = \frac{pd \cdot \Pr(H_1)}{pd \cdot \Pr(H_1) + (1 - pd) \cdot \Pr(H_0)}.$$

This makes explicit that the multiplicity adjustment acts through the prior-odds factor $\Pr(H_1)/\Pr(H_0)$.

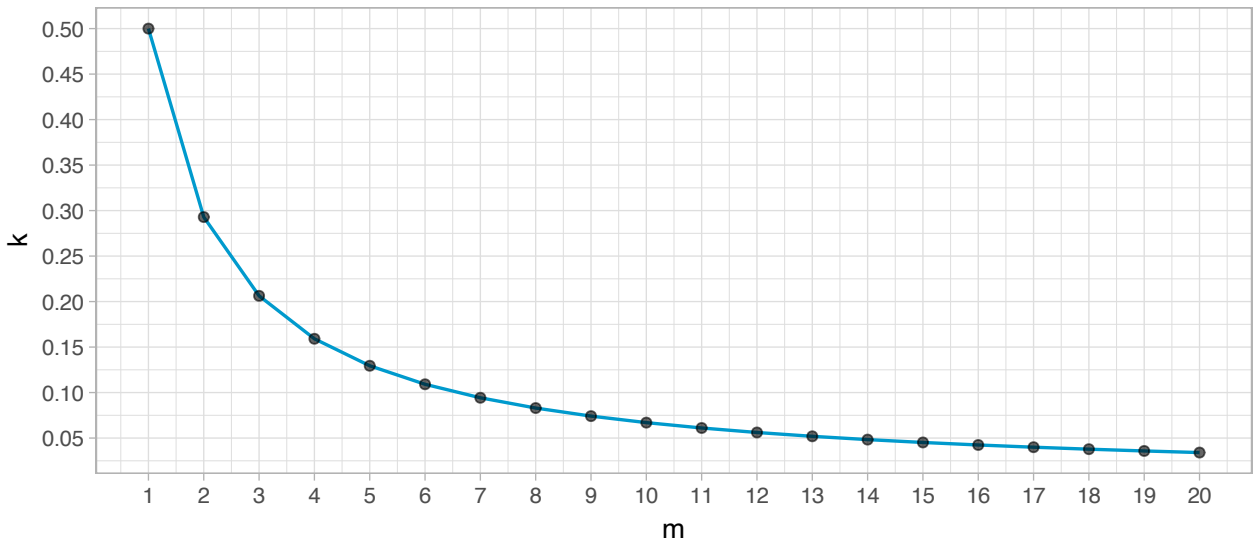
One common formalization introduces a "global null" event Q (all null hypotheses are true) and its complement $\neg Q$ (at least one null is false), and assigns equal prior probability $\Pr(Q) = \Pr(\neg Q) = \frac{1}{2}$ (Jeffreys, 1938; Westfall et al., 1997). Under independence and a common per-hypothesis probability k that a null is false, this implies

$$(1 - k)^m = 0.5 \quad \Rightarrow \quad k = 1 - 0.5^{1/m}.$$

As m increases, the per-hypothesis prior probability $k = \Pr(H_1)$ decreases (Figure Figure 5), encoding increasing prior skepticism about any particular deviation from the null; correspondingly, $\Pr(H_0)$ increases.

Figure 5

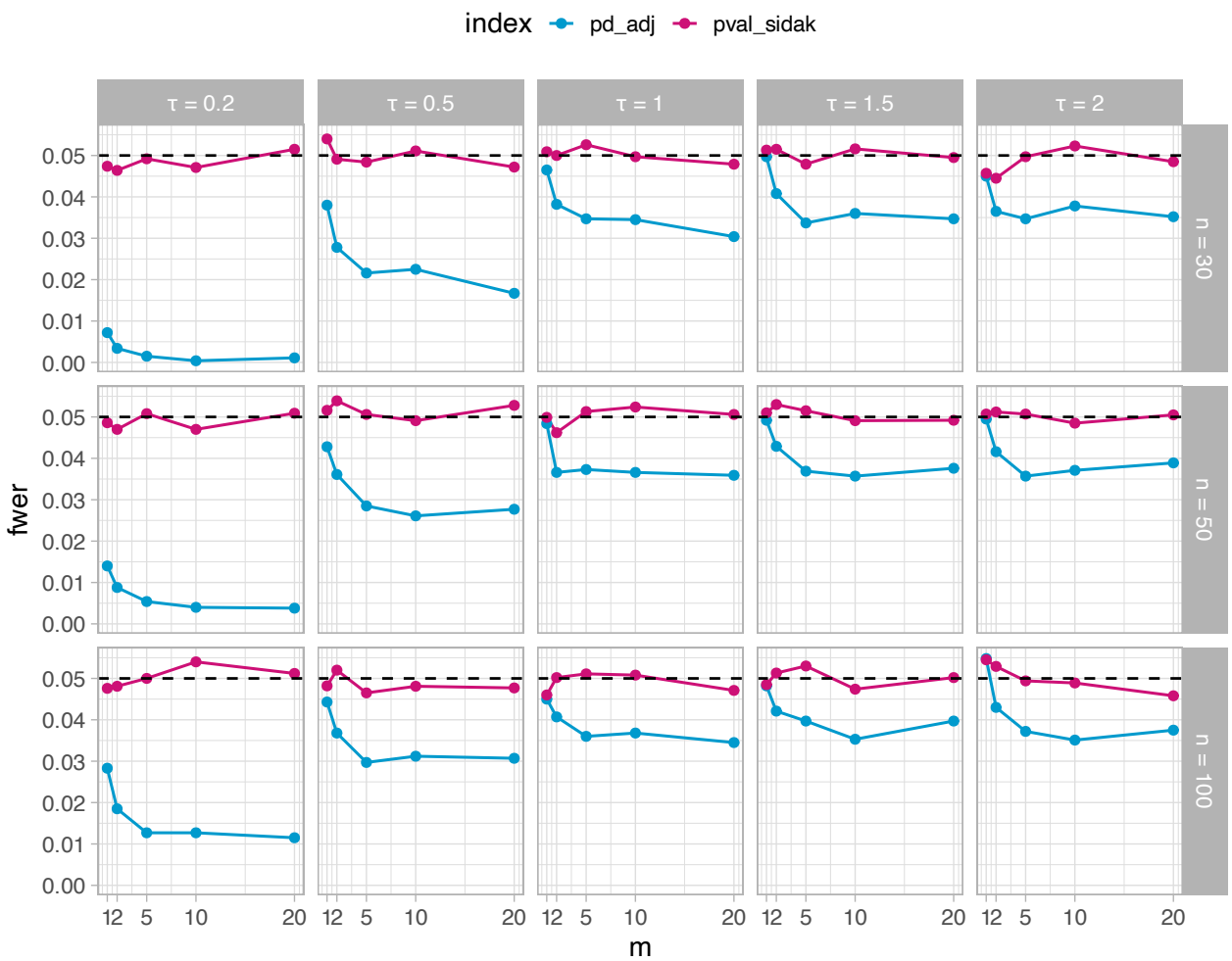
Per-hypothesis prior probability k s a function of the total number of hypotheses m, with the overall prior probability of at least one effect fixed at $q = 0.5$. The per-hypothesis prior is given by $k = 1 - 0.5^{1/m}$.



Although derived from a Bayesian prior-odds argument, the behavior of the resulting procedure closely resembles frequentists multiplicity corrections (Westfall et al., 1997). In this simulation, we compare this method with the Šidák adjustment ($p_{adj} = 1 - (1 - p)^m$), which gives exact frequentist family-wise error rate control under independence (Figure 6).

Figure 6

Familywise error rates under the global null for adjusted p-values and probability of direction. The figure shows the estimated familywise error rate (fwer), defined as the probability of at least one false positive across m tests, as a function of the number of tests (m), based on 10,000 replications under the global null ($\theta = 0$ for all tests). The red curve corresponds to declaring at least one discovery when at least one two-sided one-sample z-test p-value < 0.05, whereas the blue curve corresponds to declaring at least one discovery when at least one pd > 0.975 under the conjugate normal Bayesian model. Rows vary the per-test sample size ($n = 30, 50, 100$), and columns vary the prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$).



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Choosing the global null probability

The framework for multiplicity adjustment via adjusted prior odds naturally allows researchers to incorporate substantive prior information based on background knowledge. One approach is to specify a global prior probability that all effects are null, $q = \Pr(\text{all null})$, and translate that belief into the per-hypothesis prior probability k. Under the assumption of independence, each hypothesis is non-null with probability k, such that $\Pr(\text{all null}) = (1-k)^m$. Domain knowledge guides the choice of q: if one expects that no real effects are present, q may be large (e.g., 0.5), whereas studies targeting well-established effects might justify a smaller q (e.g., 0.1). This leads to the general calibration:

$$(1-k)^m = q \quad \Rightarrow \quad k = 1 - q^{1/m}.$$

However, with small m and low q values, this formula can paradoxically inflate rather than reduce posterior probabilities. This occurs when the per-hypothesis prior probability of the null hypothesis ($\Pr(\text{one null}) = q^{1/m}$) falls below 0.5, effectively favoring the alternative hypothesis a priori. Because we conceive this adjustment as a safeguard against inflated false positives, it is crucial to ensure it never acts in the opposite direction. To this end, the procedure should include a safeguard: if $\Pr(\text{one null}) < 0.5$, the original posterior probabilities should remain unmodified.

```
#' Prior-odds adjustment for Probability of Direction (pd)
#'
#' Adjusts a vector of Probability of Direction (pd) values using a global
#' prior probability that all tested hypotheses are null, q. The adjustment
#' converts q into a per-hypothesis prior probability H0 = q^(1/m),
#' where m is the family size, and then reweights each pd accordingly.
#'
```

```r
prior_adj = function(pd, q = 0.5, m = length(pd)) {

  stopifnot(
    "`pd` must be numeric"        = is.numeric(pd),
    "`pd` must be in [0.5, 1]"    = all(pd >= 0.5 & pd <= 1),
    "`q` must be a single number" = length(q) == 1L && is.finite(q),
    "`q` must be in (0, 1)"       = (q > 0) && (q < 1),
    "`m` must be >= 1"            = length(m) == 1L && is.finite(m) && (m >= 1)
  )

  H0 = q^(1 / m)
  H1 = 1 - H0

  if (H0 >= 0.5) {
    pd_adj = (pd * H1) / (pd * H1 + (1 - pd) * H0)
  } else {
    warning("Pr(H0) < 0.5; returning unadjusted pd")
    pd_adj = pd
  }

  return(pd_adj)
}
```

Once q is fixed, the prior odds for each individual hypothesis become a deterministic function of the family size m: larger families imply smaller k, and therefore higher prior probability toward the null for each test.

To illustrate this behavior, we simulated a mixed data-generating process in which the

global null holds with probability $\Pr(\text{all } \theta = 0) = 0.5$, otherwise $\theta = 0.25$. We repeated the previous simulation procedure with $\tau = 2$, $n \in \{30, 50, 100\}$, and analysis priors $q \in \{0.50, 0.40, 0.30, 0.20, 0.10\}$.

Across all values of m, Šidák-adjusted p-value keep the family-wise error rate (FWER; Figure 7) close to the nominal level, but this comes with a power cost (Figure 8) that becomes more pronounced as the family size increases. In contrast, the adjusted *pd* procedure attains higher power for smaller q (i.e., when the prior places less mass on the global null), but this gain is accompanied by inflated FWER.

Notably, setting $q = 0.4$ yields FWER and power trajectories that closely track the Šidák adjustment across all m, suggesting that, under this calibration, the Bayesian adjustment can effectively emulate classical strong FWER control.

Figure 7

Familywise error rate for adjusted p-value and probability of direction as a function of the number of tests (m), and prior $\Pr(\textit{all } \theta = 0)$. *The red curve corresponds to declaring at least one discovery when at least one two-sided one-sample z-test p-value < 0.05, whereas the blue curve corresponds to declaring at least one discovery when at least one pd > 0.975 under the conjugate normal Bayesian model.*

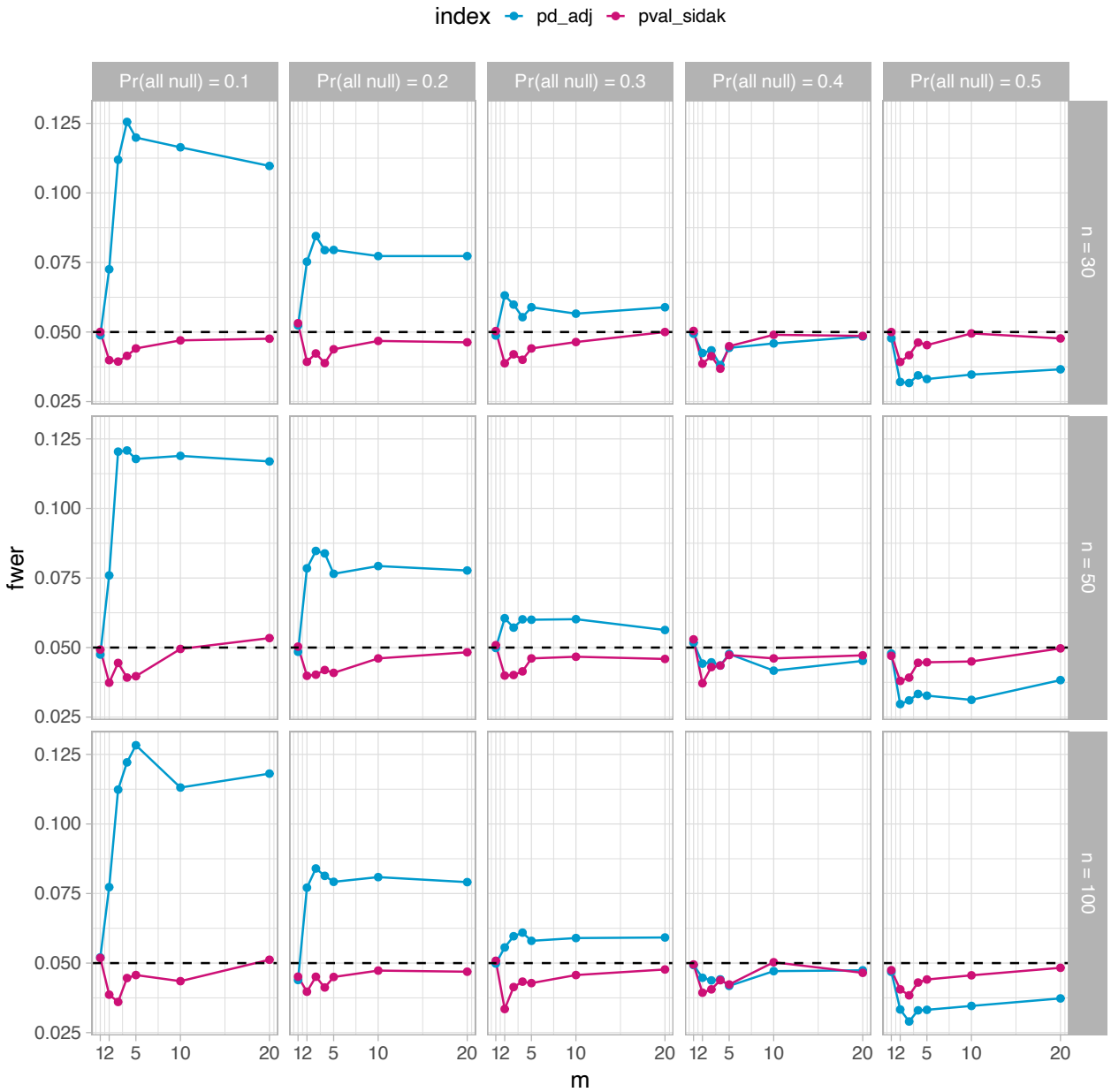
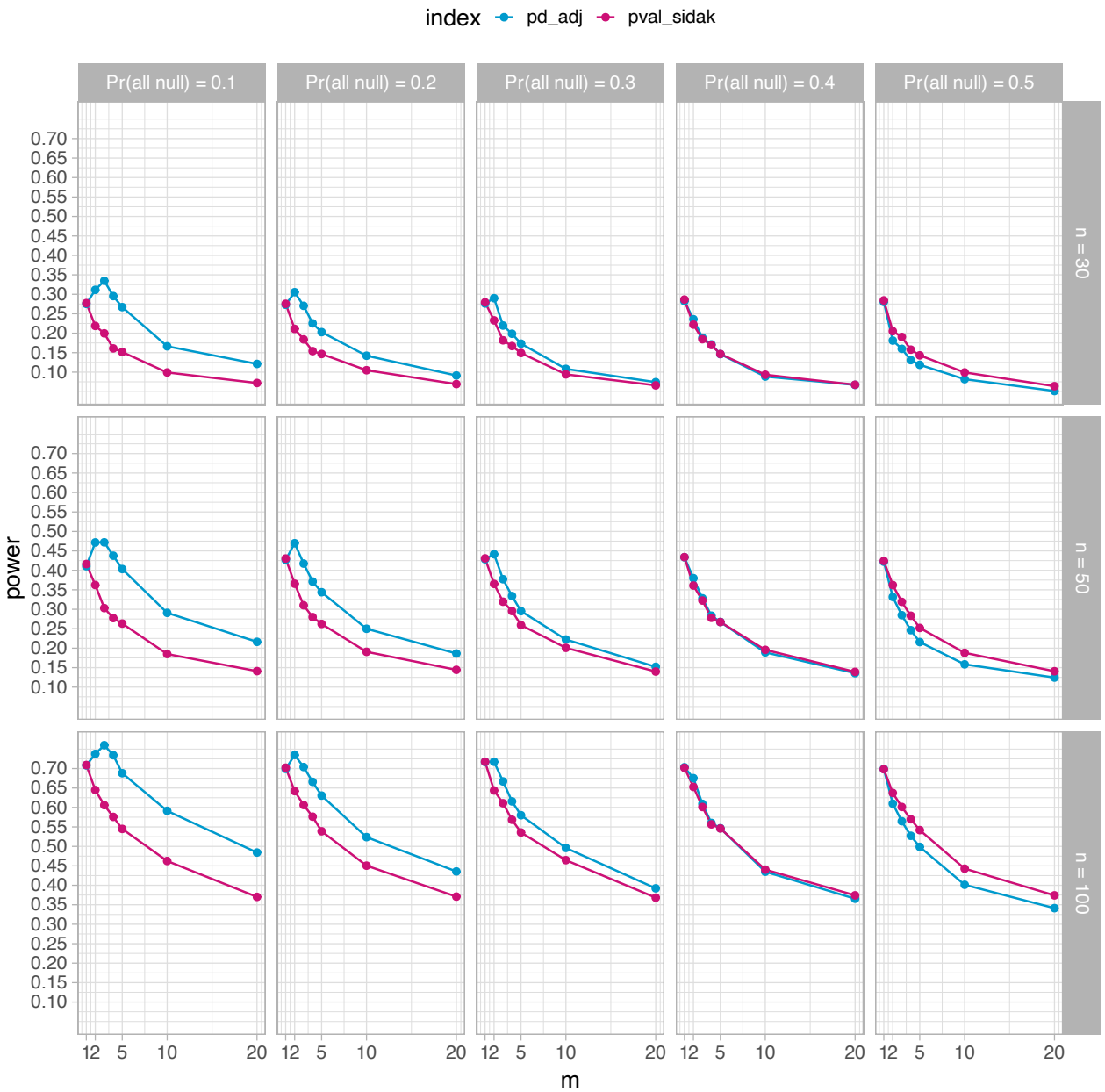


Figure 8

Power for adjusted p-value and probability of direction as a function of the number of tests (m), and prior $\Pr(all\ \theta = 0)$. *The red curve corresponds to declaring at least one discovery when at least one two-sided one-sample z-test p-value < 0.05, whereas the blue curve corresponds to declaring at least one discovery when at least one pd > 0.975 under the conjugate normal Bayesian model.*



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Discussion

The proposed multiplicity adjustment can incorporate researcher-specific priors that may draw on evidence from previous studies, theoretical constraints, measurement reliability, or structured expert elicitation [**REFs**]. For transparency, researchers should report the raw *pd* for each parameter alongside at least one multiplicity-aware quantity derived from the prior-odds adjustment. When possible, presenting both the conservative benchmark (i.e., pd when $\Pr(\text{all } \theta = 0) = 0.4$) and an informed adjustment helps clarify how sensitive conclusions are to the assumed multiplicity structure.

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Because strict independence-based adjustments can be overly conservative when hypotheses are correlated, a practical option is to replace the nominal number of tests m with an effective number m_{eff} that reflects the dependence structure. This can be done by computing a correlation matrix R capturing correlation among tests (e.g., posterior distributions from the same model) and mapping R to m_{eff} via eigenvalue-based methods (Cheverud, 2001; Del Giudice, 2021; Li & Ji, 2005). Importantly, the resulting m_{eff} will generally depend on how R is defined and estimated, so it should be treated as a sensitivity choice rather than a uniquely determined quantity. More generally, researchers can also specify hypothesis-specific values (or "classes" of hypotheses) with different null prior probabilities, for example, distinguishing target tests from exploratory ones (Guo & Heitjan, 2010).

A key advantage of this procedure is that it can be applied to *pd* computed from posterior distributions obtained under different models. In principle, the most coherent approach is to analyze all data relevant to a study within a single joint model that represents dependence explicitly, and to base inference on the resulting joint posterior distribution [**REFs**]. In practice, however, this is not always feasible, and researchers may therefore report *pd* across several analyses within the same paper.

Software implementation

For implementation, it might be best to integrate the adjustment directly into bayestestR, since it already promotes p_direction. We could reach out to its developer to discuss adding this feature instead of creating a separate package.

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